



From raw news streams to mathematically sound argumentation frameworks

uv stack

Pydantic v2

DuckDB Cache

Dung 1995 Semantics

⚡ End-to-End Data Pipeline & LLM Caching Architecture

1 News Ingestion

RSS / Atom feeds or search APIs retrieved via decoupled NewsSource adapters with topic pre-filtering.

DuckDB: news:rss [TTL]

2 AI Extraction

Pydantic-AI Agent extracts claims & attacks into typed Pydantic models. Strictly forbids URL hallucination.

DuckDB: framework:key

3 Formal Inference

Mathematical evaluation of Preferred Extensions according to Dung (1995) via backtracking or SAT solvers.

DuckDB: extensions:naive

4 Topology & Metrics

Graph-theoretic scores (acceptance rate), dilemma axes ( $A \leftrightarrow B$ ), and in/out degrees.

In-Memory (Pure Functions)

5 AI Synthesis & Export

Second Pydantic-AI Agent synthesizes perspectives into concise theses. Automated timestamped export.

output/YYYYMMDD\_\*.md

📰 Real-World Pipeline Scenario: "4-Day Work Week (8 Arguments Analyzed)"

1. EXTRACTED ARGUMENTS (A)

**A1 (Health & Focus):** Cuts burnout and sick leave; raises hourly cognitive productivity.

**A2 (Cost Shock & Inflation):** Equal pay spikes unit labor costs, threatening business viability.

**A3 (AI & Efficiency):** AI automation and streamlined workflows fully offset 20% fewer hours.

**A4 (Talent Magnet):** Attracts top talent, lowers employee turnover and costly hiring cycles.

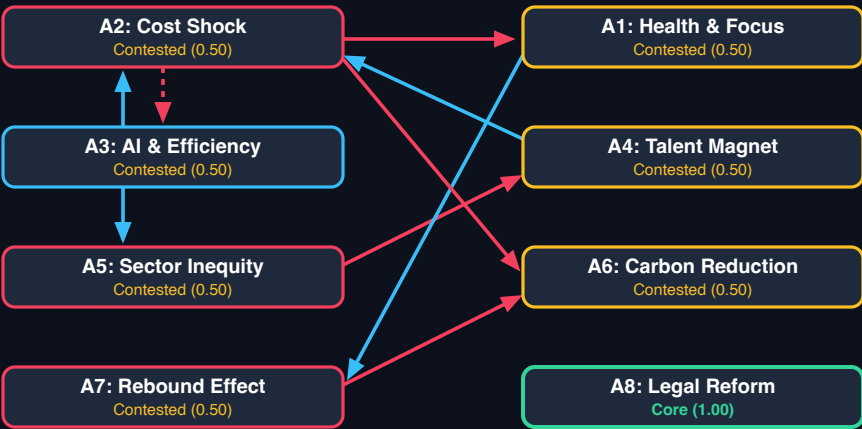
**A5 (Sector Inequity):** Unfeasible for shift/care work, creating an unfair two-tier divide.

**A6 (Carbon Reduction):** Cutting one commute day reduces office energy and transport emissions.

**A7 (Rebound Emissions):** Extra leisure triggers carbon-heavy travel and recreational spending.

**A8 (Legal Reform):** Labor laws must be modernized to enable flexible, outcome-based work.

2. ARGUMENT GRAPH & ATTACKS (R)



A2 attacks A1 & A6 | A3 & A4 attack A2 | A2 ↔ A3 Dilemma | A5 attacks A4 | A3 attacks A5 | A7 attacks A6 | A1 attacks A7 | A8 Core

3. COMPUTED PERSPECTIVES

Perspective 1 (Pro-Transformation & Climate): {A1, A3, A4, A6, A8}

AI Thesis: AI efficiency (A3) and talent retention (A4) offset reduced hours and burnout costs (A1), while commute cuts (A6) lower emissions under modernized laws (A8).

Perspective 2 (Cost Realism & Inequity): {A2, A5, A7, A8}

AI Thesis: Cost shocks (A2) and shift limits (A5) risk market inequality, while leisure rebounds (A7) diminish climate gains. Legal reforms (A8) must allow sector flexibility.

**Core Consensus:** {A8} (Score: 1.00) – Modernized labor law is accepted across all perspectives.

**Dilemma Axis:** A2 ↔ A3 – Automation capacity vs. physical frontline limits.

🤖 Dual-Stage LLM Integration

**Pydantic-AI Powered:** Stage 1 extracts unstructured news into typed ArgumentationFramework graphs. Stage 2 takes mathematically proven extensions and generates natural-language meta-theses.

uv run clashpy --solver naive

📋 Structural Roles

Mathematical classification by acceptance score across computed extensions:

Core Consensus (Score = 1.0)

Contested (0 < Score < 1)

Excluded / Rejected (Score = 0)

🔒 Zero-Leakage Secrets

Fully automated API key resolution without hardcoding:

- os.environ
- Local .env file
- Encrypted OS Keyring (db.syst.datahub)

